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ORIGINAL ARTICLE

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Insight on molecular pathogenesis and pharmacochaperoning potential in phosphomannomutase 2 deficiency, provided by novel human phosphomannomutase 2 structures

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Abstract

Phosphomannomutase 2 (PMM2) deficiency, the most frequent congenital disorder of glycosylation (PMM2-CDG), is a severe condition, which has no cure. Due to the identification of destabilizing mutations, our group aims at increasing residual activity in PMM2-CDG patients, searching for pharmacochaperones. Detailed structural knowledge of hPMM2 might help identify variants amenable to pharmacochaperoning. hPMM2 structural information is limited to one incomplete structure deposited in the Protein Databank without associated publication, which lacked ligands and residues from a crucial loop. Here we report five complete crystal structures of hPMM2, three for wild-type and two for the p.Thr237Met variant frequently found among Spanish PMM2-CDG patients, free and bound to

Alvaro Briso-Montiano and Francisco del Caño-Ochoa contributed equally to the work.
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the essential activator glucose-1,6-bisphosphate (Glc-1,6-P_2). In the hPMM2 homodimer, each subunit has a different conformation, reflecting movement of the distal core domain relative to the dimerization cap domain, supporting an opening/closing process during catalysis. Two Mg^{2+} ions bind to the core domain, one catalytic and one structural. In the cap domain, the site for Glc-1,6-P_2 is well delineated, while a Cl^- ion binding at the intersubunit interface is predicted to strengthen dimerization. Patient-found amino acid substitutions are non-homogeneously distributed throughout hPMM2, reflecting differential functional or structural importance for various parts of the protein. We classify 93 of 101 patient-reported single amino acid variants according to five potential pathogenetic mechanism affecting folding of the core and cap domains, linker 2 flexibility, dimerization, activator binding, and catalysis. We propose that ~80% and ~50% of the respective core and cap domains substitutions are potential candidates for pharmacochaperoning treatment.

KEYWORDS
congenital disorder of glycosylation type 1, human mutations, inborn errors, Jaeken syndrome, PMM2-CDG, protein structure, structure-function correlations, X-ray crystallography

1 | INTRODUCTION

Congenital disorders of glycosylation (CDGs) are defects in the building of the glycan chains of glycoproteins and glycolipids.[1-3] Phosphomannomutase 2 (PMM2; EC 5.4.2.8; Uniprot O15305) deficiency (PMM2-CDG; MIM #212065)[4-6] is the most common CDG. Over 900 cases have been recorded.[7] Estimated incidences range from 1/20 000 in the Dutch and Danish populations[8] to 1/77 000 in Stonia[9] and to 1/287 000 in Turkey.[10] The nature of the most prevalent mutations also varies geographically, as exemplified by comparing the mutational spectrum found in Spain and Portugal relative to the ones in other European countries.[11]

PMM2-CDG is a recessively inherited disorder. It manifests in childhood or infancy with failure to thrive, hypotonia and compromised function of several organs (cerebellum, peripheral nerves, liver, pericardium, ocular manifestations).[7] There also are milder, less generalized presentations.[7] The deficient enzyme, PMM2, converts mannose-6-phosphate to the mannose-1-phosphate precursor of the GDP-mannose used as high-energy mannose donor for glycosylation.[2,5]

Human PMM2 (hPMM2) is an essential enzyme. Lack of its activity is lethal, as shown in knockout mice[12] and as inferred from the lack of homozygosity for well characterized null alleles (eg, p.R141H).[13] The deficiency of hPMM2 is not compensated by human PMM1 (hPMM1), a PMM isoenzyme encoded by a paralogous gene,[14] despite the fact that hPMM1 catalyzes the same reaction as hPMM2.[5,12,15,16]

Presently there is no cure for PMM2-CDG. Several potential therapeutic options are being pursued[17,18] but only the repurposed drug acetazolamide is in clinical trial for PMM2-CDG (www.clinicaltrialsregister.eu identification number 2017-000810-44). An obvious therapeutic approach for PMM2-CDG seeks to increase residual hPMM2 activity using pharmacochaperones able to favor proper folding or to slow inactivation/degradation of pathogenic hPMM2 variants.[19] By testing a 10 000-compound library we identified eight compounds with hPMM2 pharmacochaperoning potential,[19] selecting one (CVIII: 1-(3-chlorophenyl)-3,3-bis (pyridin-2-yl) urea) for chemical optimization. Others have exploited the use of the inert β form of Gluc-1,6-P_2 as a lead compound for stabilizing pathogenic variants of hPMM2.[20]

More than 135 pathogenic hPMM2 variants, most of them single amino acid substitutions, have been found in PMM2-CDG patients (http://www.hgmdl.cf.ac.uk[21]), with two amino acid (aa) sequence variants, p.R141H and p.F119L, being the most frequent ones.[22]

Detailed knowledge of the structure of hPMM2 could be a key tool to rationalize and predict the effects of different variants, and to aid the identification of "druggable" hPMM2 variants. Unhappily, the existing structural information on hPMM2 is not optimal. It is restricted to a single structure deposited by a structural genomics group in

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the Protein DataBank (PDB; entry 2AMY; a second entry, 2Q4R derives from 2AMY, not being an independent structure), with no associated publication. This structure has no bound functional ligands and lacks aa 213-215, which belong to a key loop (see below). hPMM1 is well characterized structurally,[23,24] but it is not an ideal model for hPMM2. The two enzymes exhibit limited sequence identity (66%),[6] imperfect overlap of their substrate specificities and kinetic parameters[25,26] and hPMM2 lacks the phosphatase activity exhibited by hPMM1 vs the essential activator of these two isozymes, glucose-1,6-bisphosphate (Glc-1,6-P_2).[25-27] Furthermore, IMP strongly enhances the phosphatase activity of hPMM1 but has no effect on hPMM2, and bisphosphonates are inhibitors of hPMM1 only.[25] Therefore, the value of extrapolations to hPMM2 from hPMM1 or other PMMs from more distant organisms is unclear.

Here, we report detailed structural studies on hPMM2 using X-ray crystallography. Our results reveal complete structures of the enzyme homodimer in the absence or presence of Glc-1,6-P_2. They also clarify activator binding to the enzyme. Our findings encompass not only wild-type (WT) hPMM2, but also its pathogenic variant p.Thr237Met (abbreviated T237M; other amino acid changes will be abbreviated similarly). This variant, frequent among Spanish PMM2-CDG patients, exhibits impaired folding and stability.[28-30] The detailed and complete description of hPMM2 structural elements, of protein mobility, ligand and ionic binding and dimerization, provides clues to interpret the effects of the reported sequence variants found in PMM2-CDG patients and to identify those variants that could potentially benefit most from pharmacochaperoning therapy.

2 | MATERIALS AND METHODS

Detailed Materials and Methods, including data on commercial providers, are given in Data S1.

2.1 | Production of hPMM2 WT and T237M variant

hPMM2 cDNA (GeneBank NM_000303.3) was PCR-cloned in pOPIN-B[31] to encode hPMM2 with an N-terminal PreScission enzyme-cleavable His_6-tag. Site-directed mutagenesis was used for introducing the T237M amino acid substitution. Both proteins were produced by autoinduction[32] in *Escherichia coli* BL21 cells and were purified by two rounds of nickel affinity chromatography separated by 4 hours dialysis/digestion with PreScission enzyme for tag removal, followed by a final preparative size exclusion chromatography (SEC) step. Yield was ~15 mg of purified protein per 1 L of culture.

2.2 | Crystallization

Crystals were obtained by vapor diffusion at 21°C in hanging drops made of 1:1 mixtures of crystallization solution (0.3-0.4 M $MgCl_2$, 24% [wt/vol] PEG3350 and 0.1 M HEPES pH 7.5) and 5 mg/mL protein in SEC buffer with or without 3 mM Glc-1,6-P_2 (from Sigma). Before freezing, crystals were cryo-protected by brief passage through mother liquor supplemented with 20% (vol/vol) glycerol.

2.3 | Data collection and structure determination

X-ray diffraction data were collected at the ALBA and ESRF synchrotrons (Table 1) and processed with XDS[33] and autoPROC.[34] Crystallographic phases were obtained by molecular replacement (PHASER[35]) using separately the core (aa 1-83, 189-212, and 216-246) and cap (aa 87-184) domains of PDB file 2AMY as search models. Final models were obtained by alternate cycles of Coot[36] building and Phenix[37] or Refmac[38] refinement using CCP4.[39] Analysis used PDBePISA[40] for interactions, and PyMOL (https://pymol.org/2/) for drawing of structural figures.

2.4 | Other techniques

Analytical SEC, determination of the absolute molecular mass of the particles in SEC elution peaks by multi-angle light scattering (SEC-MALS) and isothermal titration microcalorimetry (ITC) experiments are described in the Supplementary Materials and Methods.

3 | RESULTS AND DISCUSSION

3.1 | The hPMM2 structure

We produced and purified the recombinant WT and T237M forms of hPMM2 (246 aa) as stable homodimers in solution, judged by a combination of SDS-PAGE and SEC and SEC-MALS analysis (Figure 1A,B). Both proteins formed crystals with and without Glc-1,6-P_2 that diffracted X-rays to resolutions of ~2 Å (T237M) and 2.7-3.1 Å (WT) (Table 1). Despite many attempts, we could not crystallize the complex with CVIII, although isothermal titration microcalorimetry (ITC) data were consistent with the binding of this compound to hPMM2 (K_D ~ 9 μM) (Figure 1C). This technique also indicated the high-affinity binding of Glc-1,6-P_2 to hPMM2 (K_D ~ 1 μM) in a stoichiometry of one molecule per subunit (Figure 1D).

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TABLE 1 Crystals used for X-ray diffraction

Crystal	WT-APO	WT-Glc-1,6-P_2 (co-crystallized)	WT-Glc-1,6-P_2 (soaking)	T237M-APO	T237M-Glc-1,6-P_2 (co-crystallized)
PDB file	7O0C	7O1B	7O4G	7O5Z	7O58
Data collection					
Synchrotron	ALBA	ESRF	ALBA	ALBA	ALBA
Beamline	BL13-XALOC	MASSIF-1	BL13-XALOC	BL13-XALOC	BL13-XALOC
Wavelength (Å)	0.97912	0.96600	0.97926	0.97926	0.97926
Space group	$P6_522$	$P6_522$	$P6_522$	$P6_522$	$P6_522$
Unit cell parameters					
a, b, c (Å)	70.6, 70.6, 360.8	71.0, 71.0, 364.3	71.1, 71.1, 361.3	70.9, 70.9, 363.6	70.5, 70.5, 359.7
α, β, γ (°)	90, 90, 120	90, 90, 120	90, 90, 120	90, 90, 120	90, 90, 120
Resolution range[a] (Å)	360.80-2.80 (2.95-2.80)	61.45-3.08(3.13-3.08)	361.27-2.66 (2.79-2.66)	61.43–2.07 (2.13-2.07)	59.95-1.97(2.01-1.97)
Reflections, total/ unique[a]	237 479/13763 (38 086/2003)	203 130/10988 (9568/514)	423 990/16779 (57 442/2150)	562 617/34061 (40 480/2617)	695 155/37985 (51 592/2683)
R_{pim}[a] (%)	6.7 (55.1)	8.1 (36.4)	8.5 (58.9)	3.4 (42.2)	5.7 (72.5)
I/σ_I[a]	9.5 (1.8)	8.4 (2.2)	9.2 (1.8)	15.4 (1.8)	14.7 (1.8)
Completeness[a] (%)	96.5 (100)	99.9 (100)	100 (100)	98.3 (100)	96,4 (100)
Multiplicity[a]	17.3 (19.0)	18.5 (18.6)	25.3 (26.7)	16.5 (15.5)	18.3 (19.2)
$CC_{1/2}$[a] (%)	99.4 (77.9)	99.7 (78.4)	99.6 (62.5)	99.9 (61.0)	99.9 (75.1)
Wilson B-factor	66.32	54.03	52.66	38.37	30.67
Refinement					
Resolution (Å)	39.40-2.80	60.71-3.08	60.21-2.66	61.43-2.07	59.95-1.97
Reflections used	13 643	10 928	16 663	33 915	37 823
R-factor/ R-free (%)	21.69/26.84	22.27/26.46	22.35/27.30	20.38/22.39	18.27/22.99
rmsd from ideal					
Bond lengths (Å)	0.009	0.013	0.009	0.004	0.014
Bond angles (°)	1.138	1.521	1.033	0.687	1.215
Number of					
Polypeptide chains	2	2	2	2	2
Protein non-H atoms	3801	3867	3845	3907	3950
Solvent molecules	41	55	75	243	297
Ligated Cl^-/Mg^{2+}/ Glc1,6P_2	1/4/0	1/4/1	1/4/1	1/4/0	1/4/1
Average B-factors (Å^2)					
Protein	63.4	43.3	49.9	39.4	42.6
Cl^-/Mg−1/Mg−2	49.9/64.9/65.4	32.5/45.1/45.9	33.3/57.6/53.2	26.2/38.0/38.0	34.4/39.5/39.3
Glc-1,6-P_2	–	61.7	68.3	–	81.9
Solvent molecules	60.7	45.5	47.5	42.4	48.2
Ramachandran plot (%)					
Favored	94.79	93.96	95.83	96.67	97.08
Allowed	5.00	5.42	4.17	3.33	2.92
Outliers	0.21	0.62	0	0	0

Note: Data collection and refinement statistics.

[a]Values in parentheses are the data for the highest-resolution shell.

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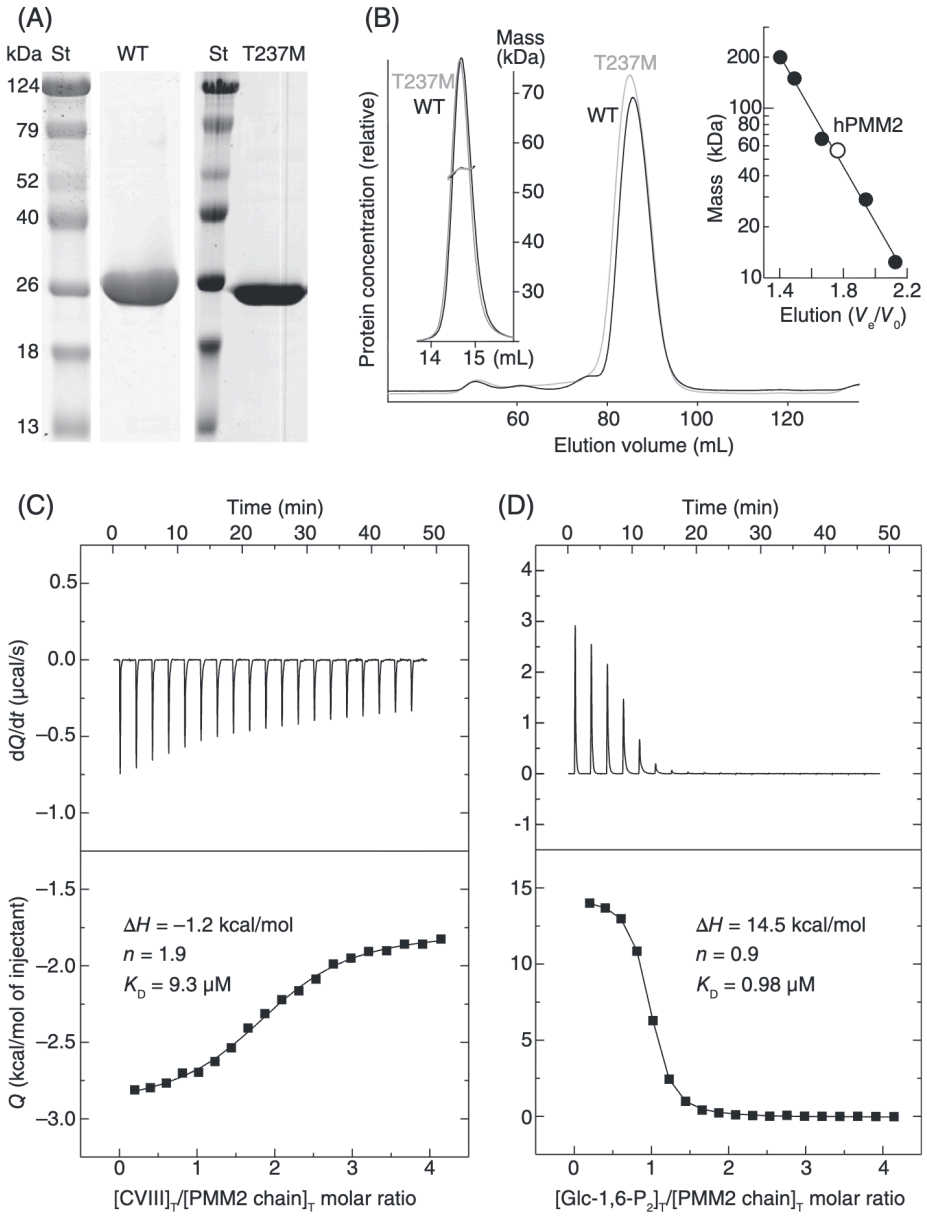

FIGURE 1 Characterization of purified recombinant hPMM2 WT and T237M variant, and ligand binding analysis. A, SDS-PAGE and Coomassie staining of purified PMM2 WT and T237M variant. Molecular weight standards (St) are indicated. B, Size exclusion chromatography (SEC) analysis. The *main figure* shows the elution profile of hPMM2 WT (black) and T237M variant (gray) in the final gel filtration step of the purification of these proteins; both proteins elute at the same position (within experimental error). *Right inset*, semilogarithmic plot of the molecular masses of protein standards (black circles) and of hPMM2 WT dimer (white circle) vs experimental elution volumes [expressed as elution volume (V_e) relative to column exclusion volume (V_o)]. *Left inset*, SEC-MALS analysis shows that hPMM2 WT (black) and T237M (gray) variant form stable homodimers in solution. The lines across the elution peaks give the absolute molecular masses (right *y* axis) determined by MALS (WT, 54.61 ± 0.38 kDa; T237M, 54.53 ± 0.38 kDa). C and D, Calorimetric titrations (ITC) of the binding to hPMM2 of compound VIII, C, and of Glc-1,6-P_2, D. Upper plots show the thermograms (thermal change as a function of time) and lower plots show the binding isotherm (normalized heat per injection as a function of the molar ligand/protein ratio). Nonlinear fitting analysis was performed as described in Data S1. The change in enthalpy (ΔH), number of estimated binding sites (n), and dissociation constant (K_D) for the ligands are indicated within the graphs. The estimated error for the ITC measurements is below 5%

The WT and T237M crystals were isomorphous and contained one enzyme homodimer in the asymmetric unit (Figure 2A). Electron density maps allowed unambiguous tracing of the entire polypeptide chain except for the four N-terminal residues (Figures 2B and S1). Irrespective of the presence or absence of Glc-1,6-P_2, the WT and T237M dimer structures were essentially identical, as shown by the low root-mean-square-deviations (rmsd)

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for their superimpositions (Table S1). However, the two subunits within the dimer showed different conformations due to changes in the position of the core domain (aa 1-83 and 189-246) relative to the cap domain (aa 87-184), which provides the entire dimerization surface (Figure 2A,C,D and Table S2). Thus, the core domain was closer to the cap

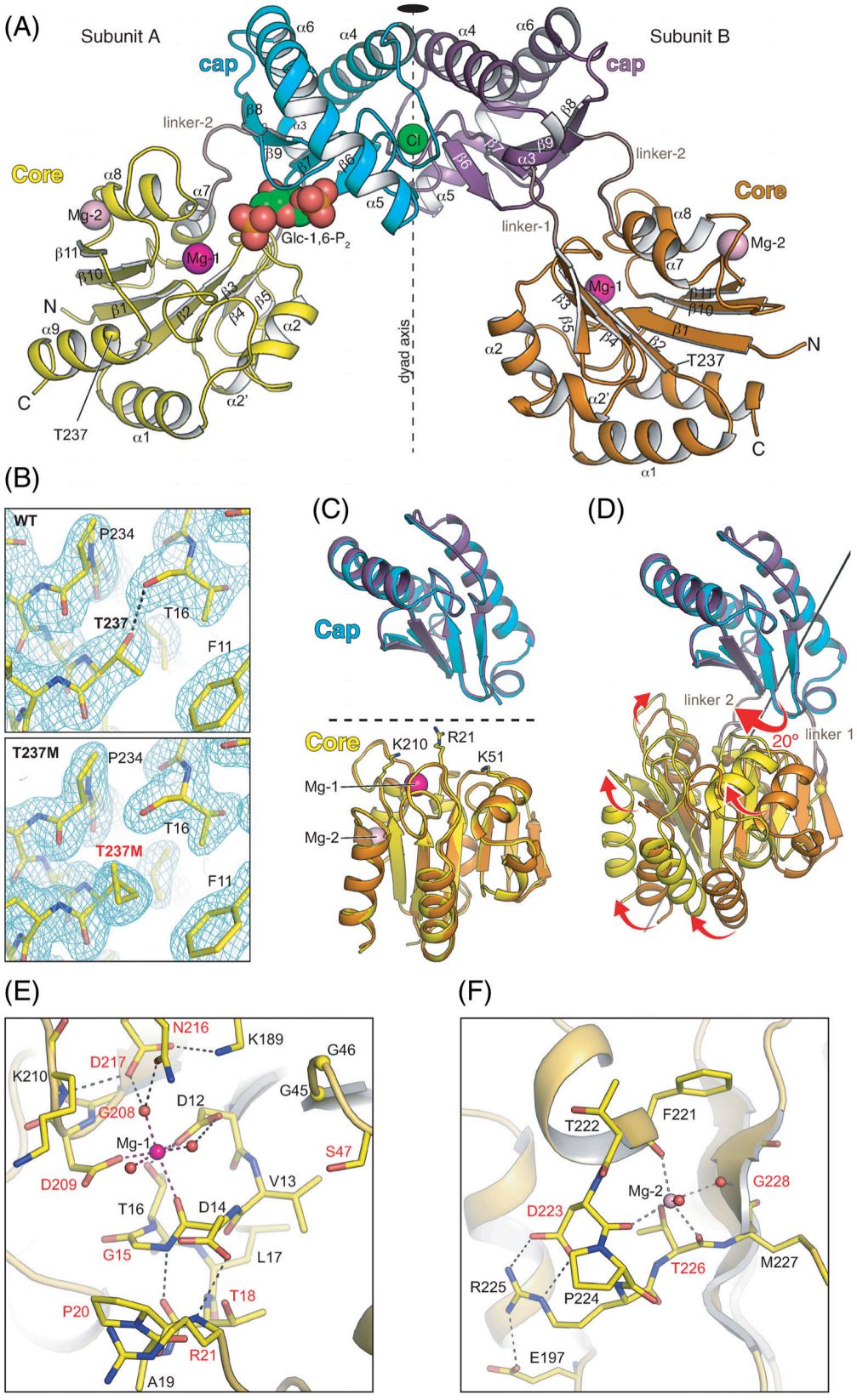

FIGURE 2 Legend on next page.

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domain in one subunit (named A) than in the other (subunit B) (Figure 2A,D). This movement reflects the flexibility of two linker regions (linker 1, aa 84-86; linker 2, aa 185-188) that connect the cap domain with the core domain, dividing the compactly folded core domain into a larger moiety (aa 1-83) and a smaller moiety (aa 189-246) (Figures 2A and S1). The observed movement narrows the distance between the two halves of the active site, which involves the cap domain for binding the Glc-1,6-P_2, as seen in our crystal structures, while the core domain provides the site for phosphoryl group transfer, as indicated in our structures by a Mg^{2+} ion bound in this domain (Figure 2A and see below). A further degree of closure than observed in our structures would be needed to properly locate the activator for occurrence of the phosphorylation reaction, as inferred from the structure of a protozoal PMM and from molecular dynamics simulations with hPMM2.[41,42]

The core domain folds as an $\alpha_4\beta_7\alpha_2$ sandwich (subscripts indicate the number of elements in each layer). Two prominent loops, β1-α1 (aa 12-24) and β10-α8 (aa 212-217), emerge from the C-edge of the mainly parallel central β-sheet, coordinating a Mg^{2+} ion (called here Mg-1) (Figures 2A,E, S1, and Table S3). This metal cation corresponds to the Mg^{2+} ion located at an equivalent site in hPMM1, believed to play a crucial catalytic role.[23] The deposited hPMM2 structure (PDB 2AMY) is incomplete, lacking this cation as well as three residues of the β10-α8 loop (^{213}PGG^{215}). The importance of Mg-1 is reflected in the distribution of clinical aa variants along the hPMM2 sequence, which shows that the Mg-1-coordinating loops are mutational hotspots in PMM2-CDG patients (Figure 3). In our structures, the core domain constantly hosts a second Mg^{2+} ion (Mg-2) that is coordinated with three residues of the α8-β11 loop (aa 223-226), two of which host mutations in PMM2-CDG patients (Figures 2A,F, 3, S1 and Table S3), supporting the importance of this cation, which possibly plays a structural role.

The core domain also hosts in its C-terminal α9 helix the Thr that is mutated to Met in the T237M variant (Figure 2A,B). The side chain of T237 makes a hydrogen bond with T16, which belongs to the Mg-1-coordinating β1-α1 loop. The loss of this hydrogen bond possibly explains the decreases in stability and activity of the T237M variant.[43] Nevertheless, the crystal structures of this variant with and without Glc-1,6-P_2 do not show important differences relative to the corresponding structures of WT hPMM2.

Each cap domain, folded as a sheet of two β-hairpins covered on one side by four α-helices, dimerizes with the other one via mutual antiparallel interactions of strand β4, helix α4, and loop α4-β4 (Figure 4A,B). The interaction surface covers 910 $Å^2$ per subunit and involves a central hydrophobic patch (I96, I100, L104, I107, F119, I120, and F122) and peripheral polar interactions (E93-R116 salt bridge; Q97-L112, N101-K115, T118-I120, and T118-H_2O-F122 hydrogen bonds) (Figure 4B). A Cl^- ion sits at the dyadic axis between both cap domains (Figures 2A and 4A), occupying a central position in the dimerization interface and making H-bonds with the NH groups of F119 and I120 from both subunits. Judged from calculations made using the PISA server,[40] this interfacial Cl^- strengthens intersubunit interactions by 22% (ΔΔG −13 kcal/mol). The importance of a stable dimer for enzyme activity of hPMM2 was previously supported experimentally,[44] and is confirmed by the clustering of disease-associated mutations in the intersubunit region, including two mutations that affect F119 and I120, the two residues that coordinate the Cl^- (Figure 3).

In the crystals of the WT and T237M forms of hPMM2 soaked with Glc-1,6-P_2 or grown in the presence of it, this activator was found binding to the cap domain of subunit A (Figures 2A and 4A,C), sitting on the inner face of the β-sheet. On the basis of current models for catalysis by PMMs,[23,42] Glc-1,6-P_2 should be able to bind in two orientations, either with the 1-phosphate or the 6-phosphate

FIGURE 2 Crystal structure of human PMM2. A, Cartoon representation of hPMM2 dimer. The cap and core domains are colored respectively in cyan and yellow in one subunit and violet and orange in the other. The position of residue T237 is indicated. Mg-1 and Mg-2 are shown as dark and light magenta spheres. The interfacial Cl^- ion is shown as a green sphere. Glc-1,6-P_2 is shown in spheres with carbon atoms in green and oxygen and phosphorus atoms red and orange. B, Detail of the $2F_{obs}$–F_{calc} electron density map at 1.0 σ in a region around T237 in the WT structure (top panel) or around the substituting methionine in the T237M structure (bottom). The dashed line represents a hydrogen bond missing in the mutated protein. C, Superposition of the isolated cap and core domains of subunit B over the corresponding domains of subunit A to show the invariance of the internal folds of these domains. The side chains of R21, K51, and K210 are shown in sticks representation to visualize that they extend towards the cap domain facing the bound Glc-1,6-P_2 (not shown). D, Superposition of the cap domains of A and B subunits reveals the movement of the core domain relative to the same domain in the other subunit (movement from B to A indicated with red arrows; the axis around which the movement can be described as a rotation is also shown). E, Detailed view of the Mg-1 at the active site, and, F, of Mg-2 bound to loop α8-β11. Water molecules are shown as red spheres while Mg^{2+} ions are colored magenta. Dashed lines converging in the Mg^{2+} ions mark the identified groups of the coordination sphere. Other polar interactions near the Mg^{2+} ions are also marked with dashed lines. Residues labeled in red have been reported to host substitutions in PMM2-CDG patients

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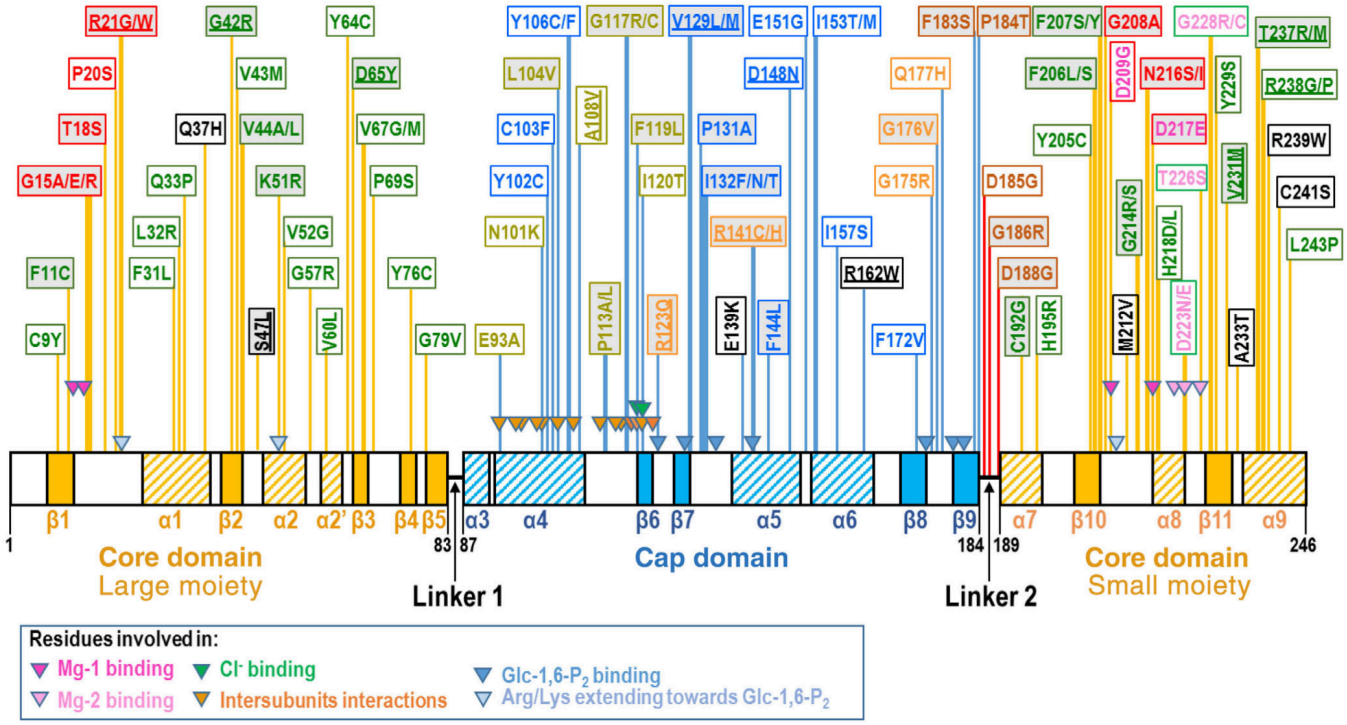

FIGURE 3 Distribution of amino acid substitutions found in PMM2-CDG patients. The bar at the bottom schematizes the enzyme polypeptide. The first moiety of the core domain, the cap domain and the second moiety of the core domain, as well as linkers- 1 and 2 are shown in different colors and labeled, with figures below the bar giving amino acid numbers at their boundaries. The α-helices and β-strands observed in our crystal structures are respectively shown as striped or solid rectangles of proportional length to the length of their sequences. They are named as in Figures 2A and S1. Banners identify the PMM2-CDG patient-reported sequence variants. The stems of the banners are colored according to the domain to which the amino acid residues pertain, with those in linker 2 colored red. Stem thickness is proportional to the number of different amino acid substitutions reported in patients for the corresponding residue (shown in the banner between/ symbols). Banners filled in gray indicate highly conserved residues in the alignment shown in Figure S2. Underlining of the amino acid substitution indicates that the nucleotide change occurred on a CpG island. The coloring of the banners (lining and name of the substitution) are green and blue for mutations predicted to primarily affect the fold of the core and cap domains, respectively; they are brown for linker 2 mutations; dark yellow for those predicted to hamper dimerization; orange for those proposed to primarily affect Glc-1,6-P_2 binding; bright red for those proposed to affect catalysis, and white for those for which the disease-causing mechanism is uncertain. For mutations possibly affecting Mg-1 or Mg-2 binding the lettering is in dark magenta and light magenta, respectively, and the banner border is bright red and green, respectively, indicating in the first case affected catalysis and in the second affected folding of the core domain. Inverted triangles identify residues involved in the functions specified in the key given at the bottom of the figure

looking towards the catalytic Mg-1 of the core domain. However, in our structures, the electron density for the activator fits best an orientation in which the 1-phosphate is anchored on the core domain via the side chains of R134, R141, S179 and the main-chain N-atoms of Q177 and I178 (Figures 4C, S1, and Table S3). The OH groups of the glucose make H-bonds with the side chains of R123, N128, S179, and D181, again from the cap domain. The 6-phosphate is less well defined, possibly because of lack of contacts with the protein, and projects towards the core domain, where three positively charged residues are exposed in the interdomain space in its direction (K210, K51, and R21) (Figure 2C). Although far away from the 6-phosphate (8-10 Å), it is possible that these residues might directly interact with the activator or with the cap domain upon further closure of the subunit.[42]

3.2 | Structure-based analysis of hPMM2 pathogenic variants

Inborn errors of metabolism offer opportunities to identify important elements and residues of the deficient enzyme, since pathogenic variants cluster in regions of high functional or structural relevance. However, in PMM2-CDG hypomorphic alleles might be overrepresented, given the absolute requirement of hPMM2 activity and the high frequency among patients of the loss-of-function R141H pathogenic variant.[13] The 101 reported (Human Gene Mutation Database, HGMD Professional 2021.2[21]) single-aa variants in PMM2-CDG patients map in 79 of the 246 residues of the hPMM2 chain (Figure 3). Many (51) of the changes affect highly conserved residues that likely play important functional or structural roles

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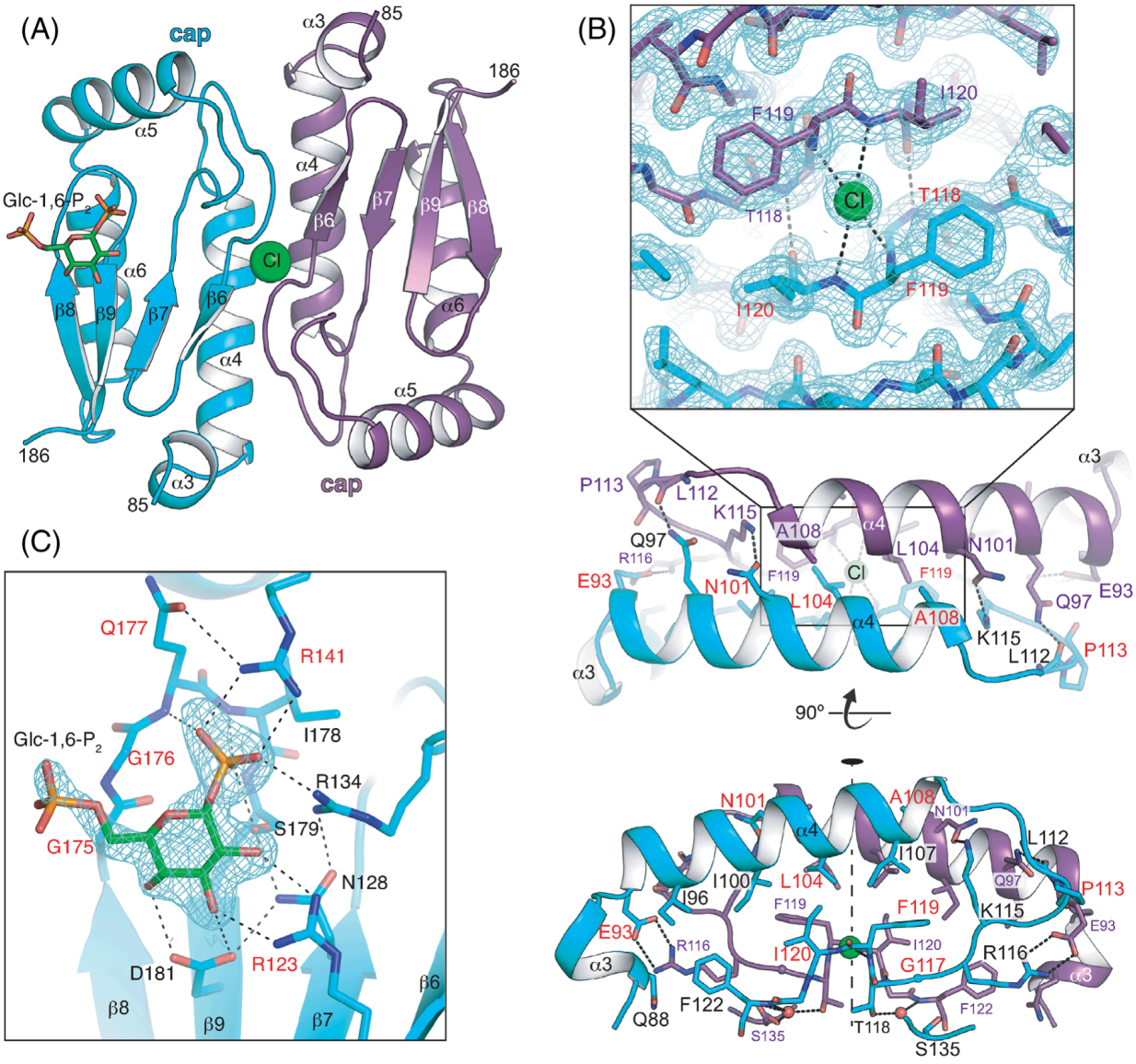

FIGURE 4 The cap domain: dimerization and activator binding. A, Ribbon representation of the cap domains in the hPMM2 dimer colored as in Figure 1. Cl^- ion is shown as a green sphere, and Glc-1,6-P_2 bound to subunit A is shown in sticks representation, with C, O, and P atoms in green, red, and orange, respectively. The dimer is viewed from the core domains along the dyadic axis, which crosses perpendicularly the image through the Cl^- ion (green sphere). B, Detail of the dimerization interface. The *top inset* shows $2F_{obs}$–F_{calc} electron density map (greenish mesh) contoured at 1.0 σ around the Cl^- ion. The view is from the core domains along the dyadic axis. *Middle and bottom panels* show two perpendicular views of the interacting elements in the dimer viewed (*middle*) along the dyadic axis from the outside of the dimer or (*bottom*) in a lateral orientation (the dyadic axis is shown as a vertical broken line). In the bottom panel, two water molecules are shown as red spheres. C, Detailed view of the interactions between the activator Glc-1,6-P_2 and the cap domain of subunit A. The mesh in blue shows the $2F_{obs}$–F_{calc} electron density map contoured at 1 σ and calculated upon refinement in absence of the activator. In panels B and C, residues hosting patient-reported mutations are labeled in red in one subunit. Hydrogen bonds and ion pairs linking both subunits or involved in the binding of Glc-1,6-P_2 are shown with dashed lines

(Figures 3 and S2). Few substitutions (18) affect the points of higher mutability represented by CpG islands (Figure 3, changes underlined). The nonhomogeneous distribution of the mutations along the hPMM2 sequence (Figures 3 and S3), further supports that many clinical sequence variants selectively pinpoint residues or regions of particular importance for the function or stability of the enzyme. Thus, substitutions appear to occur with increased frequency in the small moiety of the core domain and particularly in linker 2 (Figure S3). Linker 2 appears crucial for the expected mobility of the core domain that may accompany each catalytic cycle.[42] Substitutions within the small moiety of the core domain cluster in the β10-α8 connector loop (Figure 3) coordinating the catalytic Mg-1 (Figure 2E), and also in regions involved in Mg-2-coordination (Figures 2F and 3),

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supporting the playing of key roles by these two cations. Other variants also cluster in the carboxy-terminal end of the core domain, from β11 onwards, where misfolding could leave exposed the adjacent β1-α1 loop that is involved in catalysis/Mg-1 binding.

Based on the new structural information and with the goal of identifying hPMM2 variants amenable to pharmacochaperoning therapy, we classified 93 of the 101 patient-reported missense substitutions according to five proposed major molecular pathogenic mechanisms, while another eight substitutions are clumped in an "Uncertain effects" category (Figure 5A).

3.2.1 | Folding of the core and cap domains

Sixty aa substitutions, forty three mapping in the core domain (Figure 5B, small green spheres), and seventeen in the cap domain (Figure 5B, small blue spheres), may favor unfolding, aggregation/degradation, and subsequently, reduction of protein concentration. The majority (40) of these 60 variants may disturb the hydrophobic interactions between secondary structure elements that compose the core and cap domains, via steric hindrance or/and introducing polar groups in clusters of hydrophobic side chains (underlined in Figure 5A and shown in detail in Figure 5B). Seven substitutions in the core domain (K51R, D65Y, H218D/L T237M, and R238P/G) may have the converse effect by disturbing polar bonds that fasten together structural elements (Figure 5B, insets B.2 and B.4). R238P exemplifies a dual potential mechanism of misfolding since it should also break helix α9 by introducing therein a proline, a mechanism that would be shared by Q33P for helix α1 and by L243P for the end of helix α9. T237M could also combine two misfolding effects, disturbing structurally a cluster of hydrophobic interactions and removing a polar group involved in hydrogen bonding (Figure 2B). In addition, the missing interaction with the β1-α1 loop could also affect the coordination of the catalytic Mg-1.

Altered turns can also cause misfolding, as may be the case for the G57R, G79V, and G214R/S substitutions, which replace Gly residues in sharp turns of the core domain by larger, structurally less permissive residues (Figure 5B), while P131A and E151G do the opposite in the cap domain (Figure 5B, insets B.1 and B.3). The D148N substitution, by eliminating the negative charge that caps helix α6 at its N-end (Figure 5B, insets B.1 and B.3), might alter the orientation of this helix, causing misfolding. In the core domain, substitutions D223N/E, T226S, and G228R/C could affect the coordination of the structural Mg^{2+} ion (Mg-2) (Figures 2F and 5A,B) and thus, if Mg-2 is lost or the structural relations with it are distorted, these changes could cause improper folding.

3.2.2 | Linker 2 flexibility

Whereas linker 1 shows low conservation, all but one of the residues forming linker 2 (sequence ^{185}DGWD^{188}) are highly conserved (Figure S2). The side chains of D185 and D188 form polar bonds with the side chains of, respectively, T171 (cap domain) and Arg190 (core domain). The indole ring of W187 is encircled by and makes extensive contacts with these two domains and with linker 1, thus leaving the highly conserved Gly186 as the sole point of ample flexibility in this linker (Figure 5B, inset B.5). As already indicated, the movement of linker 2 appears crucial for the opening and closing motions needed for the catalytic cycle.[42] These motions could be compromised by mutations that increase (D185G and D188G) or restrict (G186R) the flexibility of linker 2 (Figure 5B, inset B.5, brown spheres). No substitutions affecting W187 have been reported, perhaps indicating that alterations in this crucially anchored residue may tend to be lethal for PMM2-CDG patients. Although immediately preceding this linker (Figure 3), the replacements P184T and F183S (Figure 5B, brown small spheres) also affect highly conserved residues (Figure S2) and may alter the orientation of linker 2, and thus its functionality.

3.2.3 | Dimerization interface

Ten clinical aa substitutions cluster in the region of the cap domain involved in the intersubunit interactions (α4, α4-β6 linker and beginning of β6) (Figures 3 and 4A,B). These changes (Figure 5B, yellow spheres) may weaken the intersubunit interactions and destabilize the hPMM2 dimer, highlighting the importance of dimer formation and the key role of the cap domain in this process. E93A eliminates the intersubunit ion pair with R116, while N101K should cause electrostatic clash with K115 on the other subunit, with which N101 forms a hydrogen bond (Figure 4B). A108V and G117R/C might cause steric hindrance by increasing the bulkiness of the side chains at the intersubunit interface (Figure 4B). The P113A/L substitutions may alter the sharp turn imposed by P113 to the α4-β6 connector, which is crucially involved in the intersubunit interactions (Figure 4B). The observation of early onset of severe disease but long-term survival associated with homozygosity for P113L[45] and the finding of substantial enzyme activity when this mutant form was expressed in vitro[28] indicate that the P113L substitution does not totally preclude the formation of active enzyme dimers. Disturbance of hydrophobic contacts at the dimerization interface possibly explains the pathogenicity of L104V, F119L, and I120T, which affect highly conserved hydrophobic residues (Figures 4B and S2). F119L and I120T might cause extra dimer destabilization by hampering hydrogen bonding

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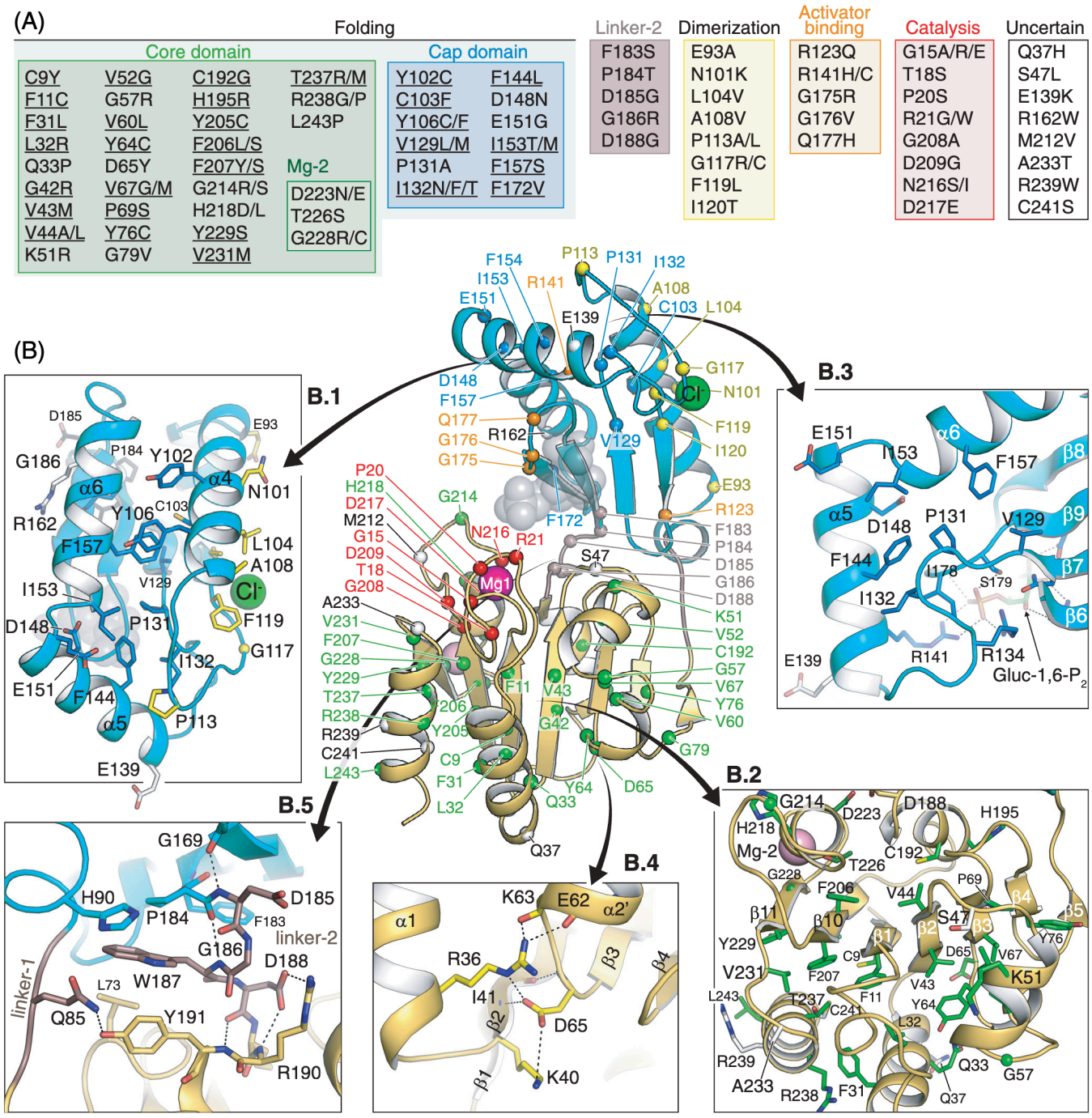

FIGURE 5 Mapping of disease-associated single amino acid substitutions on the structure of hPMM2. A, Listing of variants according to possible molecular mechanism of pathogenesis. The underlined residues are proposed to alter the clusters of hydrophobic side chains (see text). Mutations affecting Mg-2 coordination are included as folding variants of the core domain. B, Cartoon representation of hPMM2 subunit A. The cap and core domains are colored blue and yellow, respectively, and Mg-1, Mg-2, and interfacial Cl^- ion are shown as large spheres in deep magenta, light magenta and green. Glc-1,6-P_2 is shown in semitransparent gray space-filling representation. Small spheres map residues hosting patient-reported variants, colored according to the classes shown in (A): green and blue for potential misfolding changes in the core and cap domains, respectively; brown for linker 2 variants; yellow for proposed dimerization-disturbing changes; orange for activator binding site changes; red for variants potentially affecting catalysis; and white for variants of uncertain effect. Insets B.1 to B.5 zoom on specific regions to highlight detail and to show side chains of residues where substitutions have been found in patients. The color codes for these insets conform with the one for the main figure

with the Cl^- ion. Experimental studies showed that, indeed, F119L hampers dimerization but that it does not totally abolish it.[44] Compound heterozygosity for the respectively hypomorphic and inactive alleles F119L and R141H yielded hybrid dimers in which the F119L subunit exhibited normal activity.[46] However, the enzyme was

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unstable, leading to decreased PMM2 levels (documented in fibroblasts).[46] This makes patients with this most frequent disease genotype[22] potential candidates for pharmacochaperoning therapy aiming at increasing the stability of the hybrid dimer.

3.2.4 | Binding of the activator

Our structures with bound Glc-1,6-P_2 support the negative impact on Glc-1,6-P_2 binding of the substitutions R123Q, R141H/C, G175R, G176V, and Q177H (Figure 5B, small orange spheres). They affect invariant or highly conserved residues of the β6-β7 and β8-β9 hairpins of the cap domain belonging to the Glc-1,6-P_2 binding site (Figures 3, 4C, and S2). R141H and R123Q are recognized as inactivating mutations that are never found in homozygosity, but that occur frequently in PMM2-CDG patients in compound heterozygosity with an activity-decreasing but not totally inactivating second mutant allele.[22] The drastic effect of R141H could be explained by severe disturbance of a crucial close-distance ion-pair with phosphate 1 of Glc-1,6-P_2 (Figure 4C). This effect could be even more drastic for R141C. We also propose that R123Q hampers Glc-1,6-P_2 binding because of the loss of the two strong hydrogen bonds between the Arg side chain and two O atoms of the glucose. G175R and G176V may hamper Glc-1,6-P_2 binding by introducing bulky side chains at the surface under the bound activator, thus, distorting or abolishing its binding (Figure 4C). These mutations might also cause some misfolding effects resulting from the replacement by bulky residues of flexibility-endowing Gly residues near the tip of the β8-β9 hairpin on which the activator lies, and also, in the case of G176V, by exposing a hydrophobic valine side chain (Figure 5B). This last variant was found in a patient exhibiting compound heterozygosity with the relatively mild (see above) T237M change,[47] raising the question of whether G176V completely inactivated the subunit and survival was due to the partial activity of the T237M variant. The β8 to β9 hairpin is also affected by substitution Q177H, which introduces a positive charge at the hairpin tip, likely displacing the side chain of R141 from its normal position in the Glc-1,6-P_2 binding site, and possibly decreasing the affinity for the activator (Figure 4C).

3.2.5 | Catalysis

Mutations that affect residues D12, D14, and D119 possibly abrogate Mg-1 coordination and enzyme activity, given the proposed catalytic role of this Mg^{2+} ion for phosphoryl group transfer (Figure 2E).[48,49] These three residues are highly conserved in the alignment of Figure S2. D209G may be an inactivating mutation, having been found in one patient[28] in compound heterozygosity with the dimerization-impairing variant P113L that, as already mentioned, presents residual activity. The lack of reported clinical mutations affecting D12 or D14 suggests that changes at these catalytic residues are inactivating. The substitutions G15A/R/E, T18S, P20S, and R21G/W, may alter the conformation of the long convoluted β1-α1 loop (Figures 3 and 5B, red spheres), causing suboptimal positioning of the D12 and D14 catalytic residues (Figure 2E). R21 substitutions may also affect subunit closure, given the fact that R21 is one of the residues projected from the core towards the cap domain (see above). Finally, mutations G208A, M212V, N216S/I, and D217E map in the β10-α8 loop, and may affect the correct positioning of D209, and hence Mg-1 binding (Figures 2E and 5B, small red spheres).

3.2.6 | Uncertain effects

It is more difficult to propose clear-cut mechanisms for disease causation by the Q37H, S47L, E139K, R162W, M212V A233T, R239W, and C241S substitutions. All of them, excepting C241S, affect solvent-exposed residues. Q37H might disturb the last turn of helix 1, where Q37 is found (Figures 5B, inset B.2 and S1) preceded and followed by positively charged residues (R36 and K38). E139K, which replaces a negative charge by a positive charge at the beginning of helix α5, might disturb charge-capping and the positioning of this helix, although such effect is uncertain given the projection towards the solvent of the side chain of E139 in all our crystal structures (Figure 5B, insets B.1 and B.3). The S47L, R162W, M212V, and R239W substitutions possibly cause local structural rearrangements resulting from the shielding from the solvent of the novel bulky hydrophobic side chains. For example, S47 is in the β2-α2 loop (Figure 5B, inset B.2), between G46, which may be involved in the phosphoryl transfer site (Figure 2E), and D48, which caps the N-terminal dipole of helix α2 (Figure S1). Thus, the S47L substitution could cause a reorientation of the β2-α2 loop with effects on activity and/or stability. The M212V substitution maps in the β10-α8 loop (Figure 5B and Figure S1) which hosts a residue (D209) that is coordinated with Mg-1 (Figure 2E). The introduction at this position of the branched Val could modify the loop, affecting the coordination of Mg-1. The side chain of R239, in helix α9 (Figure 5B, inset B.2), makes a strong charged H-bond with D236, a residue that has its side chain hydrogen-bonded with the OH of T232 arising from the loop preceding helix α9 (not shown). The Trp of the R239W substitution cannot participate in this network of H-bonds that may be important for helix α9

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stabilization. Of the residues listed in this section, only C241 projects its side chain to the hydrophobic interior of the protein (core domain) (Figure 5B, inset B.2). The increased polarity of a Ser in the C241S substitution could disturb the hydrophobic interactions in a region densely populated by aromatic residues.

Although we have classified the variants according to their estimated major pathogenic mechanisms, it is conceivable that a given variant could have effects via several mechanisms, as exemplified by the T237M substitution. In agreement with the present predictions that such substitution should affect both folding and catalysis (via Mg-1), this mutant was found experimentally to affect both the stability and the activity of the enzyme.[28] Another example of a double effect is provided by V129M. In agreement with our prediction, this substitution was experimentally shown to decrease the stability of hPMM2.[50] However, this substitution was also reported to modestly increase K_m and K_a values for mannose-1-phosphate and mannose-1,6-bisphosphate,[50] possibly secondary to some misplacement of the neighbor Glc-1,6-P_2-interacting residue N128 (Figure 4C).

3.3 | Final considerations

The present studies exemplify with PMM2-CDG the value of detailed structural knowledge of the target protein for understanding the effects of individual aa changes found in patients of inborn errors. The present structural information, used directly or integrated into automated pathogenicity prediction servers, may be particularly important for fast evaluation of the effects of new variants now found in great numbers thanks to the extensive use of next generation sequencing. Furthermore, our structure-based assessment can be of particular relevance when a new variant is found in heterozygosity with a different aa substitution in the other allele.

An additional benefit of structure-based studies is their ability to provide clues on molecular pathogenicity mechanisms that might be amenable to drug treatment aimed at correcting the molecular derangement. Errors that decrease protein stability or that alter the dynamic equilibrium between production of well-folded and unfolded protein are typically amenable to this type of treatment. Such approach is exemplified by the use of sapropterin in phenylketonuria and by other drugs for some lysosomal storage disorders.[51-54] As already indicated, PMM2-CDG is an example of a disease for which the development of effective treatments is important, as we have a very limited repertoire of therapeutic actions to use on patients of this disease.[18]

Our present analysis of hPMM2 deficiency yielded good news, for it appears that the vast majority (~80%) of reported clinical missense changes affecting the core domain, irrespective of the moiety of that domain, and about 50% of the clinical missense mutations that affect the cap domain are predicted to hamper folding (Figure 5A), thus being potential candidates for phamacochaperoning therapy. Our structure-based inferences support our efforts towards finding compounds that could favor proper hPMM2 folding or stabilize imperfectly folded enzyme forms.[19,55] An important alley to help this approach is to guide pharmacochaperone development by structural data. Therefore, we will persist in trying to obtain hPMM2 structures in complex with promising compounds like compound VIII,[19] for which we are varying the crystallization conditions and the chemical nature of the vehicle used when adding it to the crystallization solution. We are also trying chemical modifications aimed at obtaining compounds with higher affinity and equal or better efficiency than the lead compound for increasing the residual activity of mutant forms of PMM2.

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CONFLICT OF INTEREST

Alvaro Briso-Montiano, Francisco del Caño-Ochoa, Alicia Vilas, Adrián Velázquez-Campoy, Vicente Rubio, Belén Pérez and Santiago Ramón-Maiques declare that they have no conflict of interest.

AUTHOR CONTRIBUTIONS

Santiago Ramón-Maiques: conceived the study, analyzed the data, proposed structure-based disease mechanisms, performed protein production and crystallization, collected X-ray data, solved the crystal structures and wrote the paper, read the manuscript and made contributions to improve its writing; **Belén Pérez**: conceived the study, analyzed the data, proposed structure-based disease mechanisms and wrote the paper, read the manuscript and made contributions to improve its writing;

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Vicente Rubio: conceived the study, analyzed the data, proposed structure-based disease mechanisms and wrote the paper, read the manuscript and made contributions to improve its writing; **Alvaro Briso-Montiano**: performed protein production and crystallization and solved the crystal structures, read the manuscript and made contributions to improve its writing; **Francisco del Caño-Ochoa**: performed protein production and crystallization, collected X-ray data, solved the crystal structures, read the manuscript and made contributions to improve its writing; **Alicia Vilas**: performed protein production and crystallization, read the manuscript and made contributions to improve its writing; Adrián Velázquez-Campoy performed the isothermal microcalorimetry studies, analyzed the data from these studies, draw the figures on them, helped with the writing of this part of the study, read the manuscript and made contributions to improve its writing.

DATA AVAILABILITY STATEMENT

The crystal structures and the structural factors for human phosphomannomutase 2 have been deposited in the Protein Data Bank with accession numbers 7O0C (without Glc-1,6-P_2), 7O1B (co-crystallized with Glc-1,6-P_2), 7O4G (soaked with Glc-1,6-P_2), 7O5Z (p.Thr237Met variant without Glc-1,6-P_2) and 7O58 (p.Thr237Met variant co-crystallized with Glc-1,6-P_2). Plasmids for protein expression will be provided upon request.

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